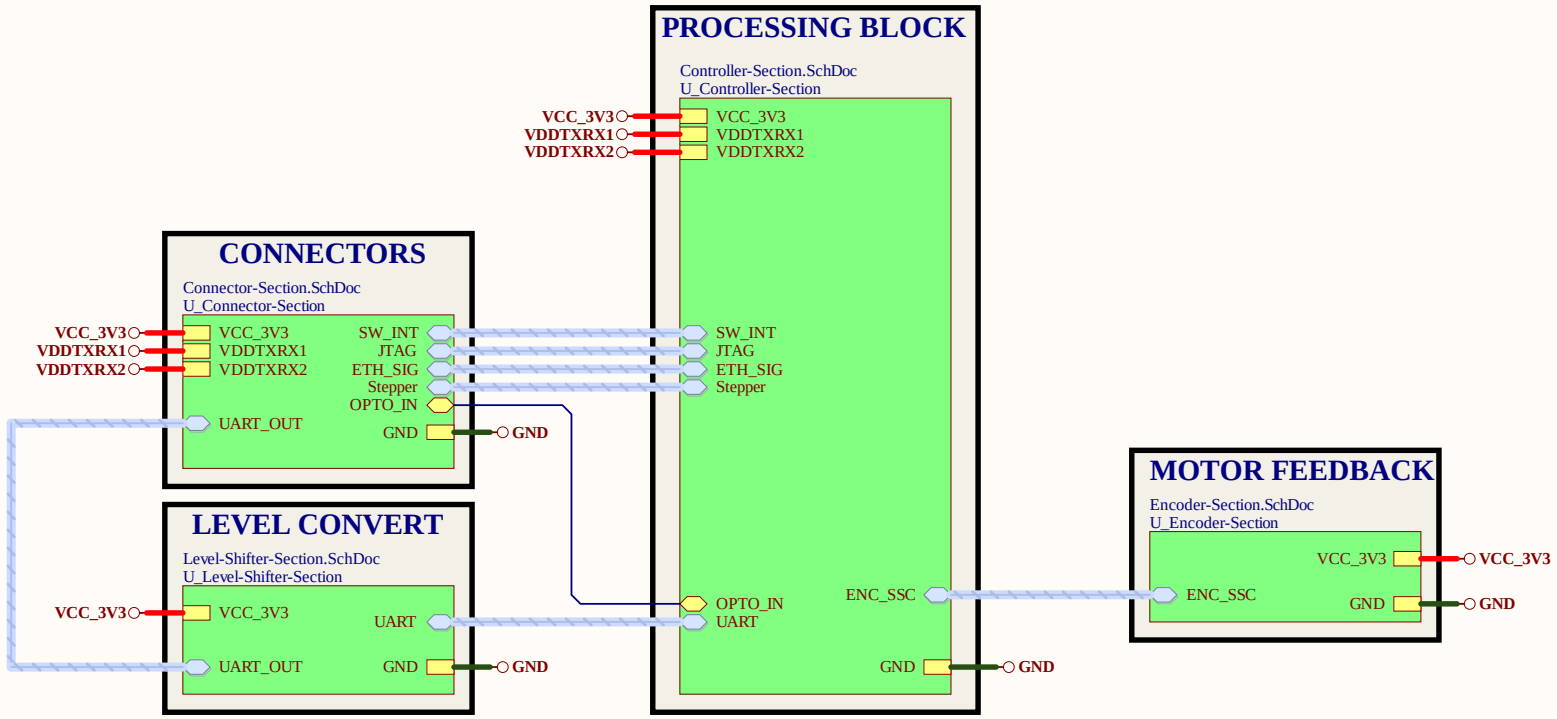
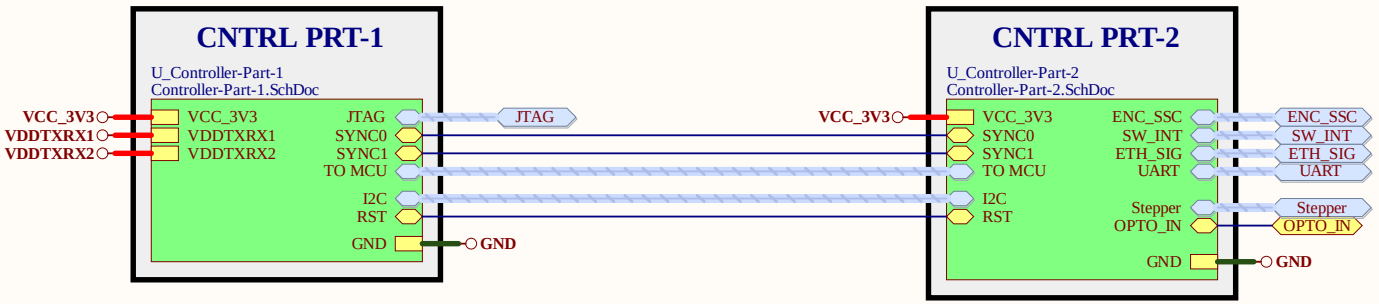
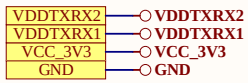
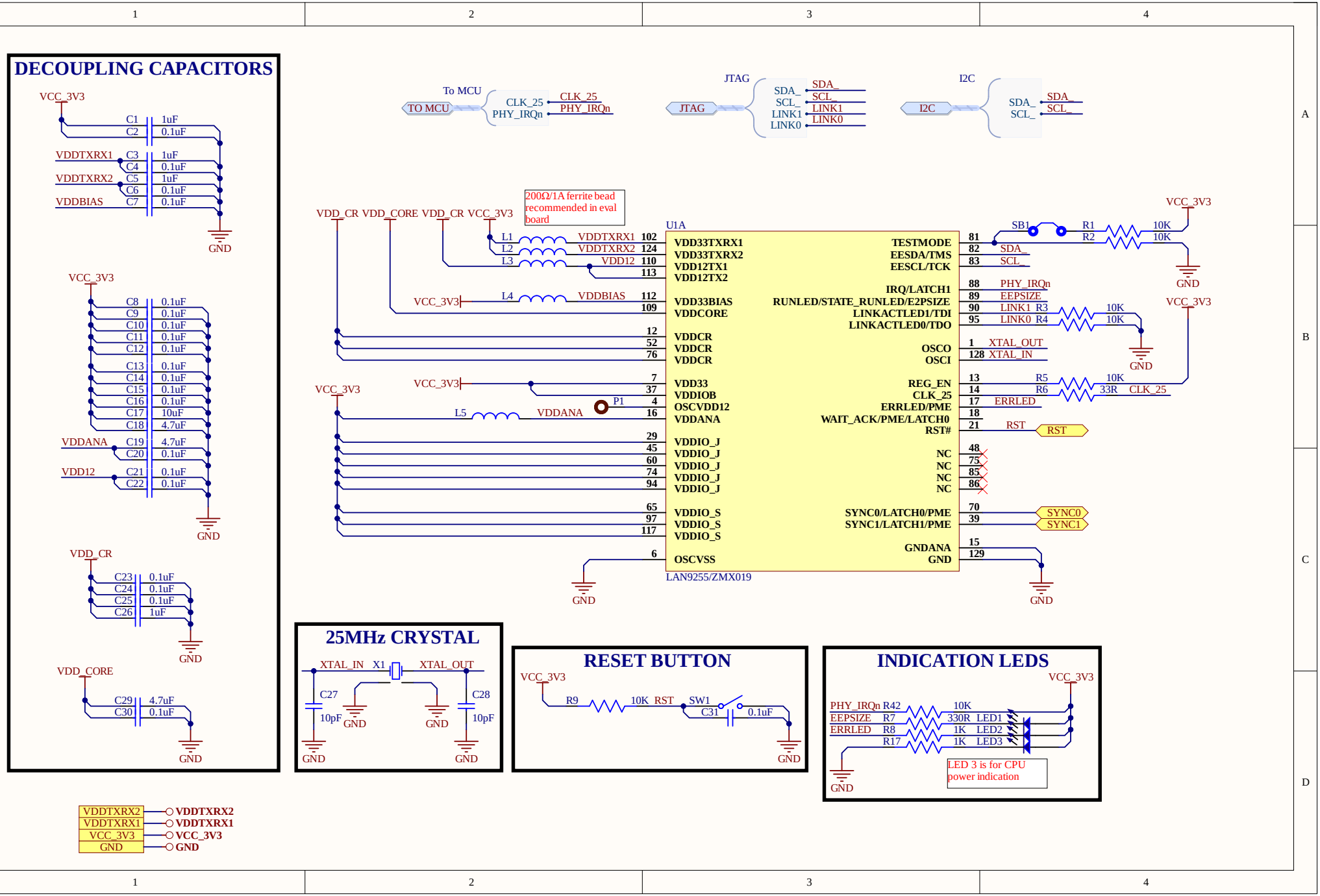
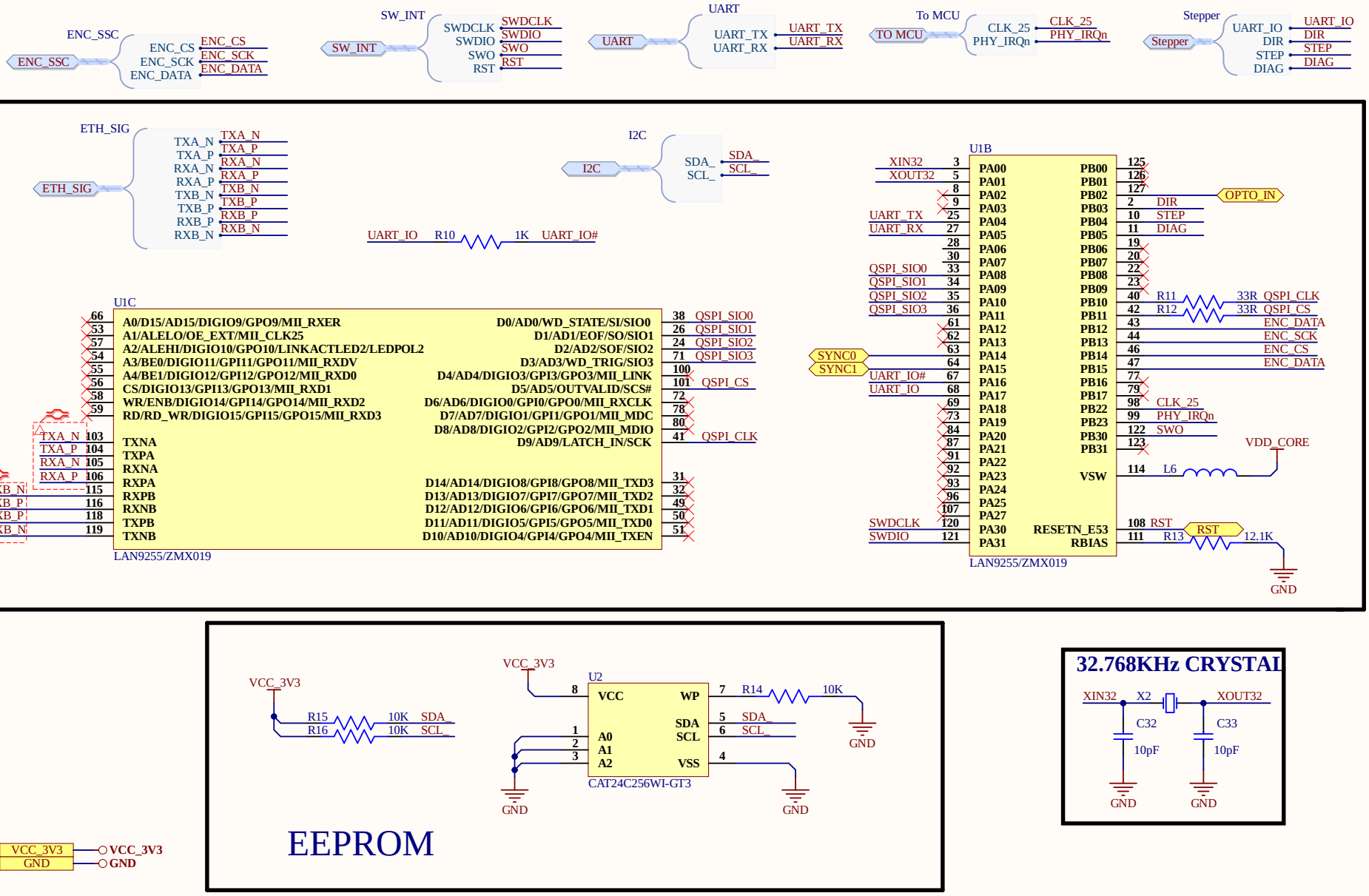

PROCESSING BLOCK
Controller-Section.SchDoc
U_Controller-Section
VCC_3V3
VDDTXRX1
VDDTXRX2
SW_INT
JTAG
ETH_SIG
Stepper
OPTO_IN
UART
ENC_SSC
GND
CONNECTORS
Connector-Section.SchDoc
U_Connector-Section
UART_OUT
LEVEL CONVERT
Level-Shifter-Section.SchDoc
U_Level-Shifter-Section
MOTOR FEEDBACK
Encoder-Section.SchDoc
U_Encoder-Section

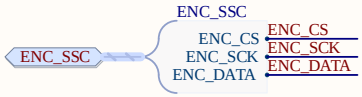

CNTRL PRT-1
U_Controller-Part-1
Controller-Part-1.SchDoc
VCC_3V3
VDDTXRX1
VDDTXRX2
JTAG
SYNC0
SYNC1
TO MCU
I2C
RST
GND
CNTRL PRT-2
U_Controller-Part-2
Controller-Part-2.SchDoc
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN


VDDTXRX2
VDDTXRX1
VCC_3V3
GND

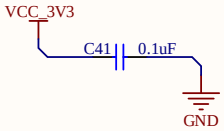

DECOUPLING CAPACITORS
VCC_3V3
C1 1uF
C2 0.1uF
VDDTXRX1 C3 1uF
C4 0.1uF
VDDTXRX2 C5 1uF
C6 0.1uF
VDDBIAS C7 0.1uF
GND
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
C12 0.1uF
C13 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 10uF
C18 4.7uF
VDDANA C19 4.7uF
C20 0.1uF
VDD12 C21 0.1uF
C22 0.1uF
VDD_CR
C23 0.1uF
C24 0.1uF
C25 0.1uF
C26 1uF
VDD_CORE
C29 4.7uF
C30 0.1uF
To MCU
TO MCU
CLK_25
PHY_IRQn
JTAG
SDA_
SCL_
LINK1
LINK0
I2C
200Ω/1A ferrite bead recommended in eval board
U1A
VDD33TXRX1
VDD33TXRX2
VDD12TX1
VDD12TX2
VDD33BIAS
VDDCORE
VDDCR
VDDIO
OSCVDD12
VDDANA
VDDIO_J
VDDIO_S
OSCVSS
TESTMODE
EESDA/TMS
EESCL/TCK
IRQ/LATCH1
RUNLED/STATE_RUNLED/E2PSIZE
LINKACTLED1/TDI
LINKACTLED0/TDO
OSCO
OSCI
REG_EN
CLK_25
ERRLED/PME
WAIT_ACK/PME/LATCH0
RST#
NC
SYNC0/LATCH0/PME
SYNC1/LATCH1/PME
GNDANA
GND
LAN9255/ZMX019
SB1
R1 10K
R2 10K
R3 10K
R4 10K
R5 10K
R6 33R
XTAL_OUT
XTAL_IN
ERRLED
RST
SYNC0
SYNC1
25MHz CRYSTAL
XTAL_IN X1 XTAL_OUT
C27 10pF
C28 10pF
RESET BUTTON
R9 10K RST
SW1
C31 0.1uF
INDICATION LEDS
PHY_IRQn R42 10K
EEPSIZE R7 330R LED1
ERRLED R8 1K LED2
R17 1K LED3
LED 3 is for CPU power indication
VDDTXRX2
VDDTXRX1
VCC_3V3
GND


ENC_SSC
ENC_CS
ENC_SCK
ENC_DATA
SW_INT
SWDCLK
SWDIO
SWO
RST
UART
UART_TX
UART_RX
To MCU
CLK_25
PHY_IRQn
Stepper
UART_IO
DIR
STEP
DIAG
ETH_SIG
TXA_N
TXA_P
RXA_N
RXA_P
TXB_N
TXB_P
RXB_P
RXB_N
I2C
SDA_
SCL_
UART_IO
R10
1K
UART_IO#
U1C
A0/D15/AD15/DIGIO9/GPO9/MII_RXER
A1/ALELO/OE_EXT/MII_CLK25
A2/ALEHI/DIGIO10/GPO10/LINKACTLED2/LEDPOL2
A3/BE0/DIGIO11/GPI11/GPO11/MII_RXDV
A4/BE1/DIGIO12/GPI12/GPO12/MII_RXD0
CS/DIGIO13/GPI13/GPO13/MII_RXD1
WR/ENB/DIGIO14/GPI14/GPO14/MII_RXD2
RD/RD_WR/DIGIO15/GPI15/GPO15/MII_RXD3
TXNA
TXPA
RXNA
RXPA
RXPB
RXNB
TXPB
TXNB
D0/AD0/WD_STATE/SI/SIO0
D1/AD1/EOF/SO/SIO1
D2/AD2/SOF/SIO2
D3/AD3/WD_TRIG/SIO3
D4/AD4/DIGIO3/GPI3/GPO3/MII_LINK
D5/AD5/OUTVALID/SCS#
D6/AD6/DIGIO0/GPI0/GPO0/MII_RXCLK
D7/AD7/DIGIO1/GPI1/GPO1/MII_MDC
D8/AD8/DIGIO2/GPI2/GPO2/MII_MDIO
D9/AD9/LATCH_IN/SCK
D14/AD14/DIGIO8/GPI8/GPO8/MII_TXD3
D13/AD13/DIGIO7/GPI7/GPO7/MII_TXD2
D12/AD12/DIGIO6/GPI6/GPO6/MII_TXD1
D11/AD11/DIGIO5/GPI5/GPO5/MII_TXD0
D10/AD10/DIGIO4/GPI4/GPO4/MII_TXEN
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
QSPI_CS
QSPI_CLK
LAN9255/ZMX019
U1B
XIN32
XOUT32
UART_TX
UART_RX
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
SYNC0
SYNC1
UART_IO#
UART_IO
SWDCLK
SWDIO
PA00
PA01
PA02
PA03
PA04
PA05
PA06
PA07
PA08
PA09
PA10
PA11
PA12
PA13
PA14
PA15
PA16
PA17
PA18
PA19
PA20
PA21
PA22
PA23
PA24
PA25
PA27
PA30
PA31
PB00
PB01
PB02
PB03
PB04
PB05
PB06
PB07
PB08
PB09
PB10
PB11
PB12
PB13
PB14
PB15
PB16
PB17
PB22
PB23
PB30
PB31
VSW
RESETN_E53
RBIAS
OPTO_IN
DIR
STEP
DIAG
R11
R12
33R
QSPI_CLK
QSPI_CS
ENC_DATA
ENC_SCK
ENC_CS
ENC_DATA
CLK_25
PHY_IRQn
SWO
L6
VDD_CORE
RST
R13
12.1K
GND
LAN9255/ZMX019
VCC_3V3
R15
R16
10K
SDA_
SCL_
U2
VCC
WP
SDA
SCL
A0
A1
A2
VSS
R14
CAT24C256WI-GT3
EEPROM
32.768KHz CRYSTAL
X2
C32
C33
10pF
VCC_3V3
GND

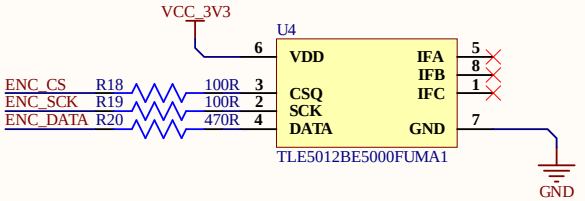
## CONTROL SIGNAL




## DECOUPLING CAPACITORS

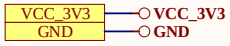



## ENCODER FEEDBACK

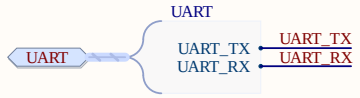



Need to add Series Resistances

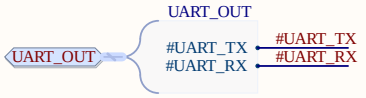
## POWER SIGNALS



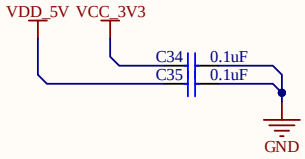
## UART SIGNALS



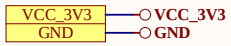

## UART OUT SIGNALS



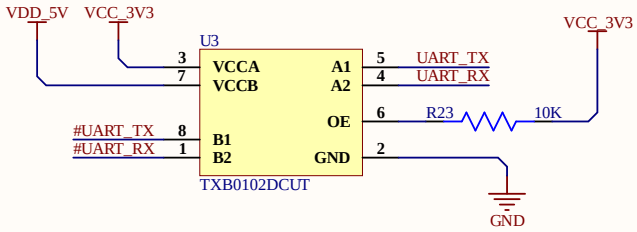

## DECOUPLING CAPACITORS




## POWER SIGNALS



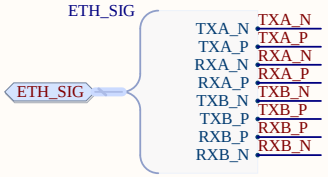

## LEVEL SHIFTER



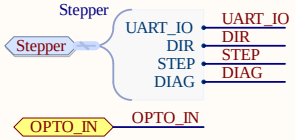

Setting the OE input high enables the device.
Datasheet: page-23

Will be changed with smaller part when library is available

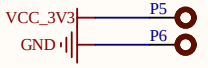
## ETHERCAT SIGNALS



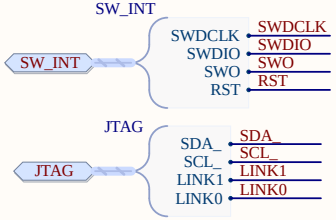

## CONTROL SIGNALS



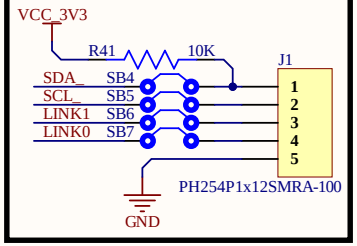

## TEST POINTS



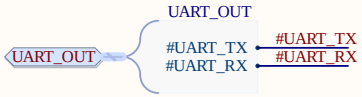

## PROGRAMMING SIGNALS



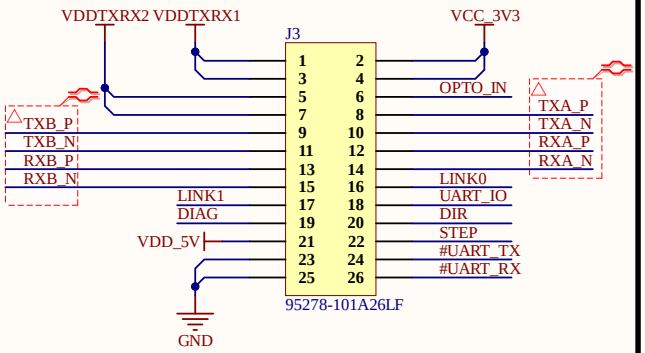

## JTAG HEADER



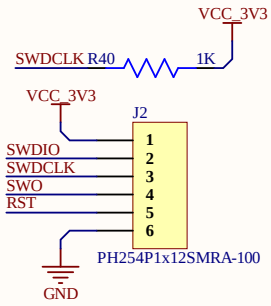

## UART OUT SIGNALS



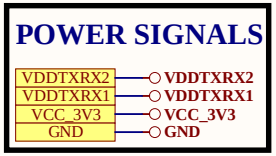

## BOARD TO BOARD CONNECTOR

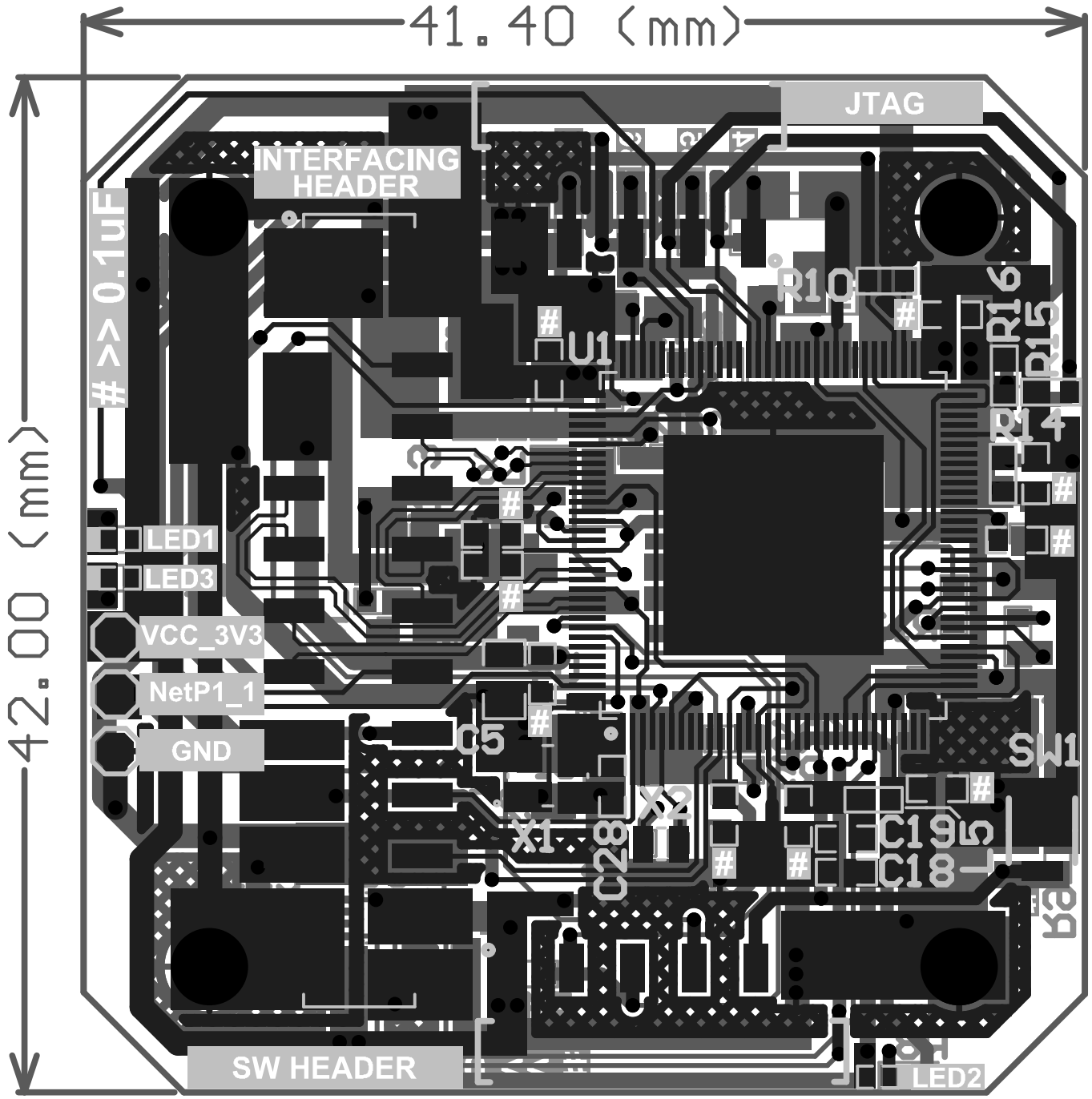



## SW HEADER




## POWER SIGNALS




41.40 (mm)
42.00 (mm)
JTAG
INTERFACING HEADER
SW HEADER
U1
R10
R16
R15
R14
C5
C28
X1
X2
C19
C18
SW1
LED1
LED3
LED2
VCC_3V3
NetP1_1
GND
0.1uF